Introduction to Scientific Writing and Data Analysis

Science

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Introduction to Scientific Writing and Data Analysis (A08041)

Short Title:	Intro Sci Writing & Analysis		
Faculty	Science and Computing		
Department	Science		
Cluster	Placement and Projects		
Author	EWOODBYRNE		
Credits	5	**Level:**	Intermediate

Description of Module / Aims

This module is designed to enable the student to read and interpret academic writing, and to produce a competently written and well-structured literature review. The student will choose a topic from a list, and will construct a literature review based on three to five papers provided by the tutor, and two to three papers sourced by the student.

Programmes

`SCIE-0061`	BSc in Horticulture (`WD_SHORT_D`)	3 / 5 / M
`SCIE-0061`	BSc in Horticulture (`WD_SHORT_DX`)	3 / 5 / M

Indicative Content

- Critical reading
- Development of academic writing skills and referencing
- Literature searching
- Structuring a literature review
- Basic statistical analysis
- Data presentation: graphical and tabular

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Interpret published academic work to identify themes and trends.
2. Analyse published academic work and comment upon its significance.
3. Construct a literature review on the current state of knowledge in a given topic area which is prepared using appropriate structure, language, grammar, punctuation and referencing.
4. Analyse data using basic statistical techniques.
5. Use appropriate software to present data in graphical and tabular formats.
6. Prepare an abstract.

Learning and Teaching Methods

- Tutorials.
- Practical training in library and/or computer laboratory.

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	
Assignment	60%	1,2,3
In-Class Assessment	20%	4,5
Assignment	20%	6

Assessment Criteria

<40%: Little or no evidence of interpretation of the supplied papers. Poorly structured and/or presented literature review. Incorrect or inadequate referencing. Little or no ability to analyse and present data. Poorly written abstract.

40%-49%: Demonstrates a basic level of understanding of the supplied papers. Shows a limited ability to identify themes/trends and to structure a literature review. Makes an attempt to reference but with errors. Some evidence of proofreading. Some ability to analyse and present data. Limited ability to produce an abstract.

50%-59%: Demonstrates a good understanding of the supplied papers. Identifies themes and trends. Attempts to paraphrase and consider opposing ideas. Referencing largely correct. Proofreading generally good. Good attempt at analysing and presenting data. Good attempt at producing an abstract.

60%-69%: Demonstrates a good understanding of the supplied papers. Identifies themes and trends. Paraphrases and presents opposing ideas competently. Few errors in referencing and proofreading. Competent data analysis and presentation. Competently produced abstract.

70%-100%: Demonstrates a high level of understanding of the supplied papers. Identifies themes and trends. Paraphrases and presents opposing arguments to a high standard of competence. Referencing correct throughout. Evidence of excellent proofreading. Excellent standard of data analysis and presentation. Excellent abstract.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Tutorial	24	
Independent Learning	111	

Essential Material

- Bell, J. and S. Waters. _Doing Your Research Project: A Guide for First-time Researchers_. 6th ed.. Berkshire: Open University Press, 2014.
- Cargill, M. and P. O'Connor. _Writing Scientific Research Articles: Strategy and Steps_. West Sussex: Wiley-Blackwell, 2009.

Requested Resources

- Room Type: Computer Lab